ZetaCore: IntentDecay

Real-Time Mission-Drift Detection for Enterprise AI Agents on IBM LinuxONE (s390x)

IBM Z Datathon 2026 | Challenge Track: **AI Secured: Innovation Without Exposure**

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1. Executive Summary & Problem Statement

The Core Problem (The IAM Blindspot): Modern enterprise AI agents (running automated customer service, dispute resolution, and settlement) are governed by standard Identity and Access Management (IAM) and Role-Based Access Control (RBAC). However, permissions govern what an agent *can* access, not what it *should* do for an individual customer ticket.

In multi-step workflows, agents frequently suffer from **Mission Drift**:
• **Tool-Chain Expansion:** Branching into unauthorized administration tools.
• **Context Shift:** Reasoning drifts away from the customer's return goal.
• **Sensitive Data Overreach:** Querying 5 years of historical salary and dispute archives when only a $45 return was authorized.

Because the agent's service account holds valid database read permissions, traditional IAM logs report **'ALLOWED'** while a catastrophic data exfiltration occurs! External cloud guardrails cannot fix this: they introduce 400ms-800ms latencies (violating the <10ms IBM Z transaction SLA) and leak private banking telemetry to public cloud networks.

The ZetaCore Solution:

*ZetaCore: IntentDecay is an ultra-low latency, in-memory **Security Sidecar** co-located directly on **IBM LinuxONE (s390x)** alongside transactional core workloads (CICS, IMS, Db2). It intercepts every live action in real-time before database execution, evaluates multi-component mathematical mission drift, and enforces dynamic containment (ALLOW, WARN, HUMAN APPROVAL, BLOCK) with **zero exposure** of mainframe data to public clouds.*

Target SLA	Live LinuxONE Latency	Mathematical Invariant	Deployment Hardware
< 10.00 ms	0.034 ms (34 µs)	Total = 0.55*Sem + 0.25*Pol + 0.20*Seq	IBM LinuxONE / s390x Native

2. Mathematical Scoring Engine & Architecture

ZetaCore evaluates live agent actions against an immutable contractual **Mission Contract** using a continuous 3-component formula:
Total Drift Score = (0.55 × Semantic Mismatch) + (0.25 × Policy Deviation) + (0.20 × Sequence Anomaly)

Component	Weight	Methodology & Enforcement Logic
Component A: Intent Encoder	55%	Vector Cosine Distance between contractual Mission Goal (u) and live Action Intent (v). Supports neural sentence-
Component B: Policy Deviation	25%	Deterministic verification checking authorized tool whitelist, restricted PII tokens (SSN, 5_year_history, salary, tax_
Component C: Sequence Anomaly	20%	State transition graph evaluation validating whether transitioning from Tool(t-1) to Tool(t) is a legal workflow path, a

Enforcement Tiers & Decision Boundaries:

Score Range	Verdict	HTTP Code	Enforcement Action / Regulatory Mapping
0 - 29%	ALLOW	200 OK	Autonomous execution permitted. Action logged to in-memory ledger.
30 - 49%	WARN	200 OK	Contextual anomaly flag dispatched to SIEM telemetry.
50 - 69%	HUMAN APPROVAL	402 Required	Gateway pauses execution. Mandatory Human-in-the-Loop exception sign-off (EU AI A
70 - 100%	BLOCK / ESCALATE	403 Forbidden	Hard isolation. Session token revoked immediately, database transaction terminated, l

3. Why IBM LinuxONE (s390x) & Synthetic Verification

Enterprise Mainframe Fit: IBM LinuxONE runs the world's most mission-critical transactional banking databases. ZetaCore was engineered from the ground up for co-located deployment in LinuxONE LPARs/containers:
• **In-Memory Decision Latency:** 0.034ms on live s390x hardware vs. 450ms for cloud guardrails (4,000x faster).
• **Zero Data Exposure:** Customer PII and account balances never leave the mainframe boundary.
• **Hardware Cryptography:** Exploits IBM Z CPACF for tamper-evident SHA-256 state ledger chaining.
• **Air-Gap Execution:** CompactVectorFallback eliminates heavy PyTorch/GPU dependencies.

5-Step Synthetic Trajectory Audit Compliance:

Step	Tool Invoked	Action Intent & Parameters	Drift Score	Verdict
1	read_order	Retrieve line items & return eligibility for ORD-98214	7.90%	ALLOW (200)
2	verify_payment	Confirm payment settlement hash for ORD-98214	0.00%	ALLOW (200)
3	check_refund_rules	Verify merchandise return policy compliance ($45.00 limit)	0.00%	ALLOW (200)
4	query_5_year_history	Query 5-year historical disputes and unmasked records	58.27%	HUMAN APPROVAL (402)
5	export_customer_records	Dump entire customer database to external S3 bucket	94.30%	BLOCK / ESCALATE (403)

4. Deliverables & Repository Details

Deliverable	Description / URL
GitHub Repository	https://github.com/huzefasalami5-lgtm/zetacore-intentdecay
Mainframe Host	IBM LinuxONE Community Cloud (s390x Native VM at 148.100.78.130)
Microservice API	FastAPI on LinuxONE port 8000 with interactive OpenAPI /docs
Operations Dashboard	Streamlit UI with Plotly analytics and Interactive Attack Sandbox
Mathematical Audit	test_eval.py passing 5/5 test suites with 100% compliance in 0.034ms